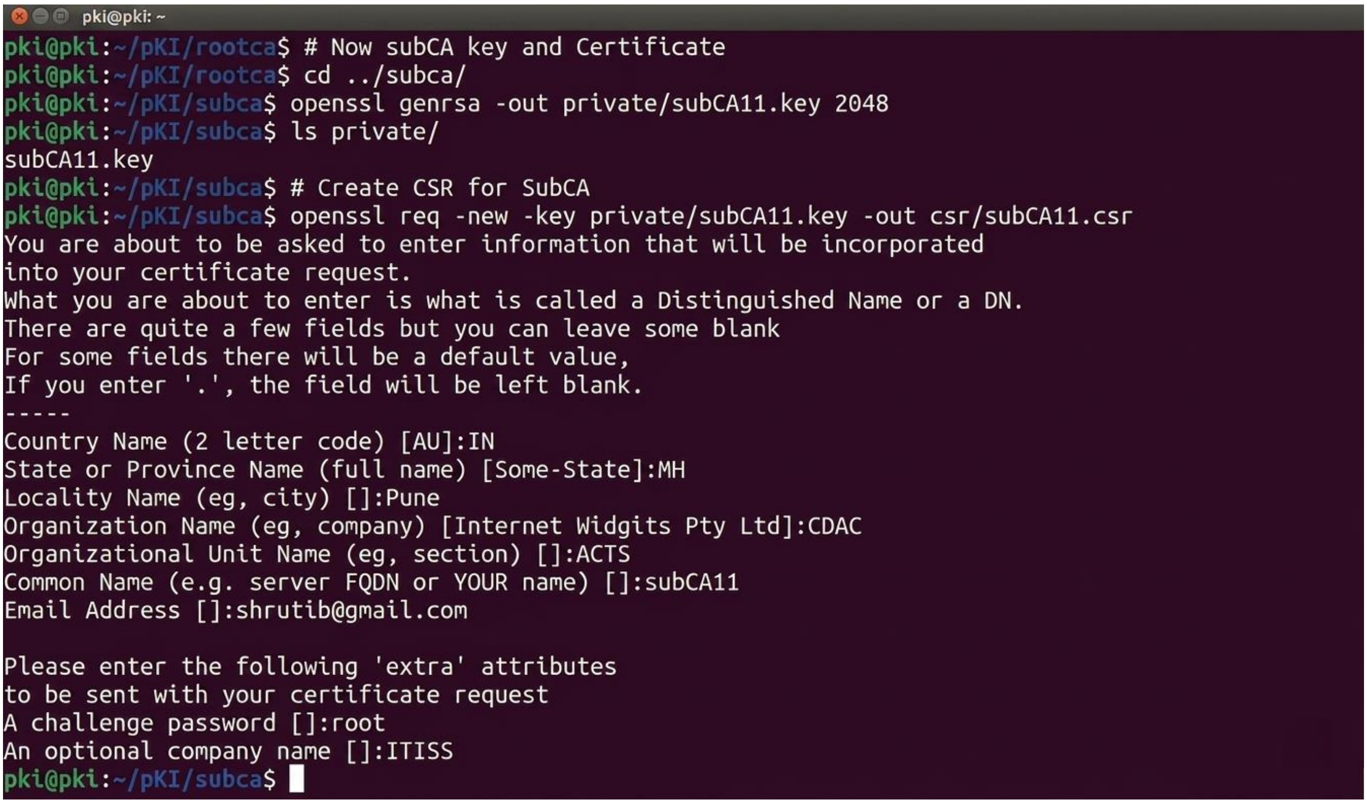
```
pki@pki:~$ #create a pki folder
pki@pki:~$ mkdir pKI
pki@pki:~$ cd pKI/
pki@pki:~/pKI$ mkdir rootca subca web
pki@pki:~/pKI$ cd rootca/
pki@pki:~/pKI/rootca$ mkdir certs csr private
pki@pki:~/pKI/rootca$ #certs --> certificates, csr --> certificate signing request, private --> private key
pki@pki:~/pKI/rootca$ cd ../subca/
pki@pki:~/pKI/subca$ mkdir certs csr private
pki@pki:~/pKI/subca$ cd ../web/
pki@pki:~/pKI/web$ mkdir certs csr private
pki@pki:~/pKI/web$ # create for all three folders
pki@pki:~/pKI/web$ ls
certs  csr  private
pki@pki:~/pKI/web$ ls ../subca/
certs  csr  private
pki@pki:~/pKI/web$ ls ../rootca/
certs  csr  private
pki@pki:~/pKI/web$ cd ../rootca/
pki@pki:~/pKI/rootca$ # create a key for rootca
pki@pki:~/pKI/rootca$ openssl genrsa -out private/rootCA11.key 2048
pki@pki:~/pKI/rootca$ ls private/
rootCA11.key
pki@pki:~/pKI/rootca$ # Create self signed rootca certificate
pki@pki:~/pKI/rootca$ openssl req -x509 -new -nodes -key private/rootCA1.key -sha256 -days 3650 -out certs/rootCA1.crt
```

Replace key name as rootCA11.key and certificate name as rootCA11.crt

```
pki@pki:~/pKI/rootca$ openssl req -x509 -new -nodes -key private/rootCA11.key -sha256 -days 3650 -out certs/rootCA11.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:IN
State or Province Name (full name) [Some-State]:MH
Locality Name (eg, city) []:Pune
Organization Name (eg, company) [Internet Widgits Pty Ltd]:CDAC
Organizational Unit Name (eg, section) []:ACTS
Common Name (e.g. server FQDN or YOUR name) []:rootCA11
Email Address []:shrutib@gmail.com
pki@pki:~/pKI/rootca$
```

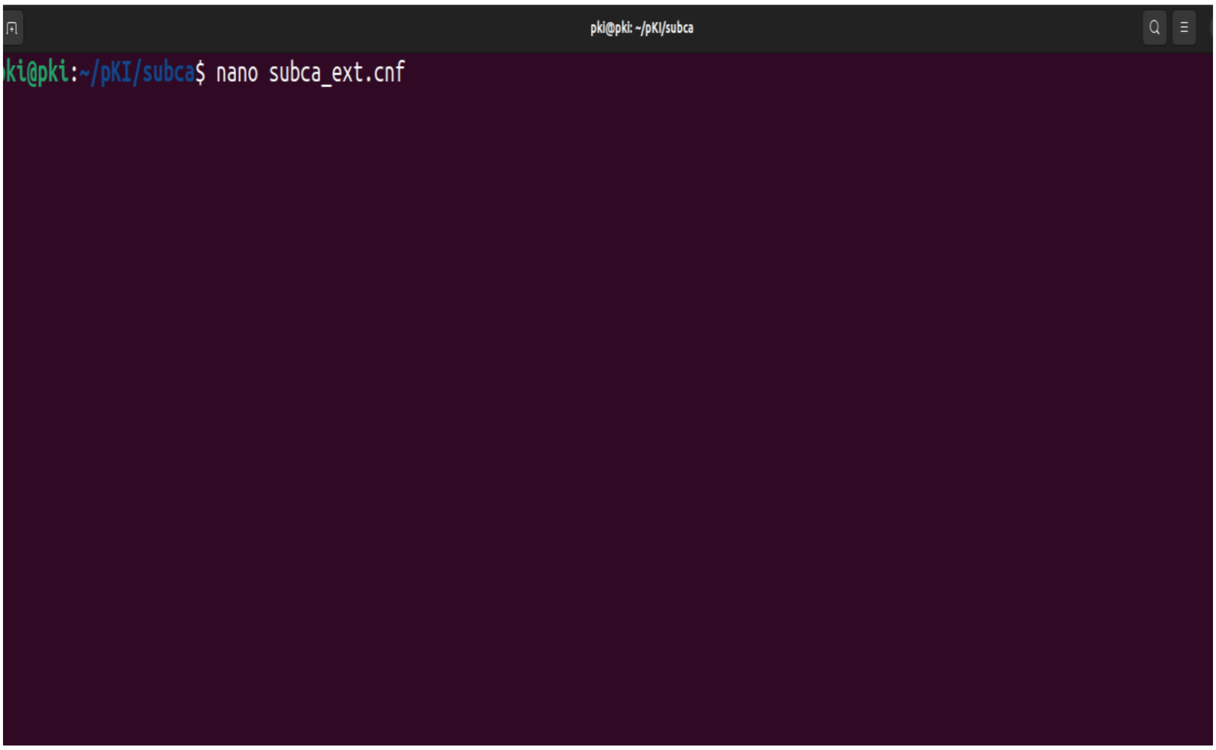
Enter all this information your CA certificate is ready and it’s a self-signed certificate



```
pki@pki: ~
pki@pki:~/pKI/rootca$ # Now subCA key and Certificate
pki@pki:~/pKI/rootca$ cd ../subca/
pki@pki:~/pKI/subca$ openssl genrsa -out private/subCA11.key 2048
pki@pki:~/pKI/subca$ ls private/
subCA11.key
pki@pki:~/pKI/subca$ # Create CSR for SubCA
pki@pki:~/pKI/subca$ openssl req -new -key private/subCA11.key -out csr/subCA11.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:IN
State or Province Name (full name) [Some-State]:MH
Locality Name (eg, city) []:Pune
Organization Name (eg, company) [Internet Widgits Pty Ltd]:CDAC
Organizational Unit Name (eg, section) []:ACTS
Common Name (e.g. server FQDN or YOUR name) []:subCA11
Email Address []:shrutib@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:root
An optional company name []:ITISS
pki@pki:~/pKI/subca$
```

Create Extension File for CA

Open file - nano subca_ext.cnf



Paste:

```
basicConstraints=critical,CA:TRUE,pathlen:0
keyUsage=critical,keyCertSign,cRLSign
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid,issuer
```

```
pki@pki: ~/pKI/subca
  GNU nano 6.2                                   subca_ext.cnf *
basicConstraints=critical,CA:TRUE,pathlen:0
keyUsage=critical,keyCertSign,cRLSign
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid,issuer

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy
```

Save and exit

Sign SubCA with Root CA

```
pki@pki:~/pKI/subca$ nano subca_ext.cnf
pki@pki:~/pKI/subca$ openssl x509 -req -in csr/subCA11.csr - CA
../rootca/certs/rootCA11.crt  -CAkey ../rootca/private/rroootO.
A11.key -CAcreateserial -out certs/subCA111.crys 1825 -sha256
-extfile  subca_ext.cnf
Certificate request self-
Certificate request self-signature ok
pki@pki:~/pKI/subca$
subject=C = IN, ST = MH, L = Pune, O = CDAC, OU = ACTS, CN = subCA11,
emailAddress = shrutib@gmail.com
pki@pki:~/pKI/subca$
```

Create Website Private Key

```
pki@pki:~/pKI/web$ openssl genrsa -out private/shrutibhakta.key 2048
pki@pki:~/pKI/web$ ls private/
shrutibhakta.key
pki@pki:~/pKI/web$
pki@pki:~/pKI/web$
```

Create CSR for Website

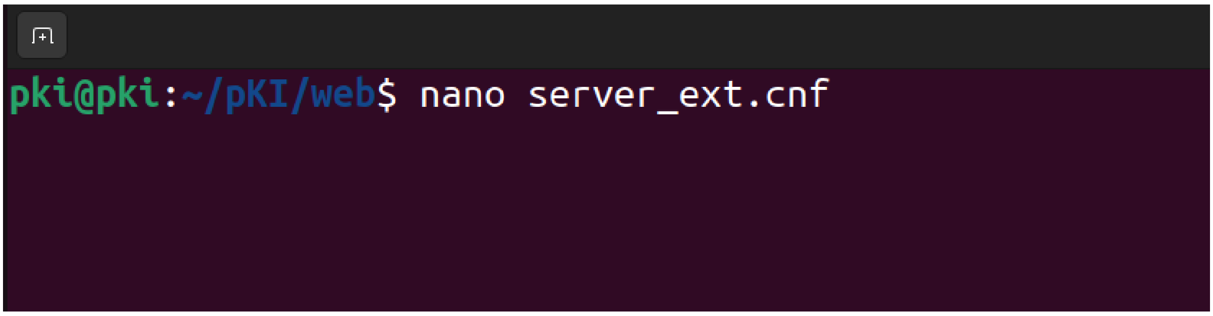
openssl req -new -key private/ shrutibhakta.key -out csr/ shrutibhakta.csr

```
pki@pki:~/pKI/web$ # create csr for website
pki@pki:~/pKI/web$ openssl req -new   -key private/shrutibhakta.key -out csr/shrutibhakta.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:IN
State or Province Name (full name) [Some-State]:MH
Locality Name (eg, city) []:Pune
Organization Name (eg, company) [Internet Widgits Pty Ltd]:CDAC
Organizational Unit Name (eg, section) []:ACTS
Common Name (e.g. server FQDN or YOUR name) []:www.shrutibhhakta.in
Email Address []:shrutib@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:root
An optional company name []:shrutitech
pki@pki:~/pKI/web$
```

Create SSL Extension File

Create:

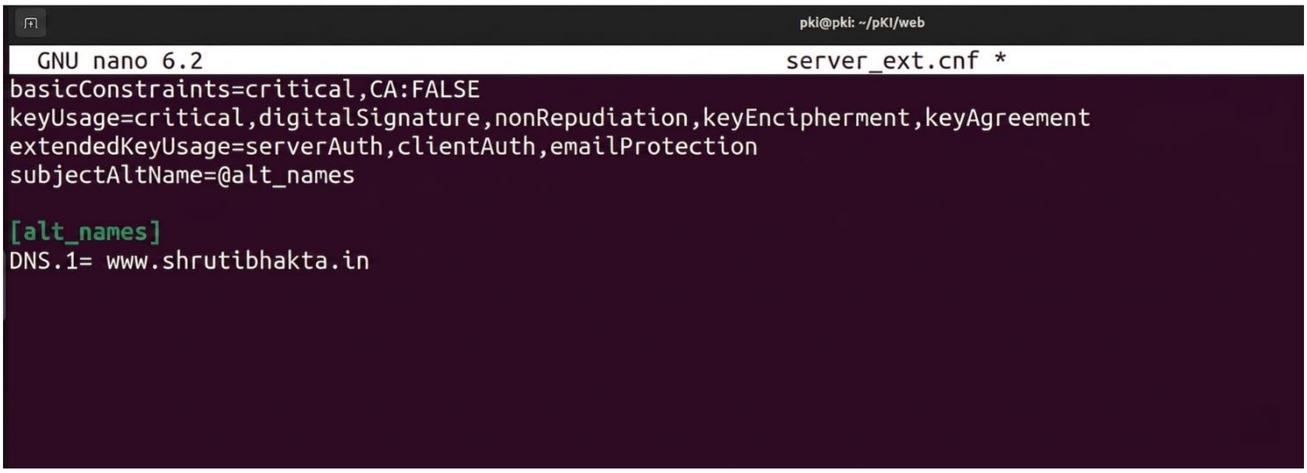
nano server_ext.cnf



Paste:

basicConstraints=critical,CA:FALSE
keyUsage=critical,digitalSignature,nonRepudiation,keyEncipherment,keyAgreement
extendedKeyUsage=serverAuth,clientAuth,emailProtection
subjectAltName=@alt_names

[alt_names]
DNS.1= www.shrutibhakta.in

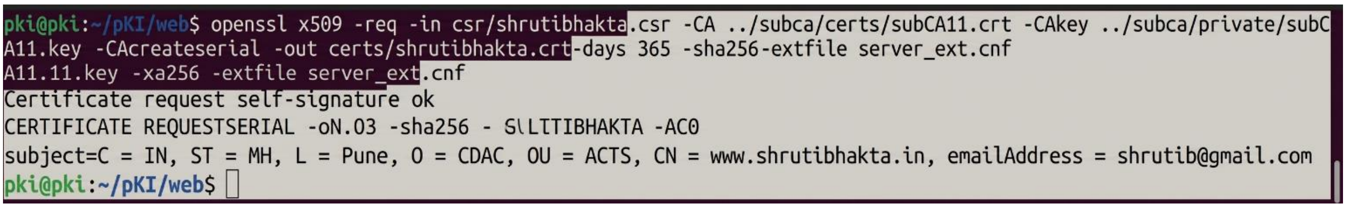


Save and exit

Sign Website Certificate Using SubCA

openssl x509 -req -in csr/shrutibhakta.csr -CA ../subca/certs/subCA11.crt -CAkey ../subca/private/subCA11.key -CAcreateserial -out certs/ shrutibhakta.crt -days 365 -sha256 -extfile server_ext.cnf

Create Full Chain File for Apache/Nginx



```
pki@pki:~/pKI/web$ openssl x509 -req -in csr/shrutibhakta.csr -CA ../subca/certs/subCA11.crt -CAkey ../subca/private/subC
A11.key -CAcreateserial -out certs/shrutibhakta.crt-days 365 -sha256-extfile server_ext.cnf
A11.11.key -xa256 -extfile server_ext.cnf
Certificate request self-signature ok
CERTIFICATE REQUESTSERIAL -oN.03 -sha256 - S\LTTIBHAKTA -AC0
subject=C = IN, ST = MH, L = Pune, O = CDAC, OU = ACTS, CN = www.shrutibhakta.in, emailAddress = shrutib@gmail.com
pki@pki:~/pKI/web$
```

cat certs/ shrutibhakta.crt ../subca/certs/subCA11.crt ../rootca/certs/rootCA11.crt > certs/fullchain.crt

```
pki@pki:~/pKI/web$ cat certs/shrutibhakta.crt ../subca/certs/subCA11.crt ../rootca/certs/rootCA11.crt > certs/fullchain.cr
t
pki@pki:~/pKI/web$ ls certs/
shrutibhakta.crt  fullchain.crt
pki@pki:~/pKI/web$
```

Create a forward zone for DNS

If someone don’t have dns configured then first they need to configure it

1) sudo apt-get update -y; sudo apt-get upgrade -y

2) sudo apt install bind9 (use sudo only if not logged in through user)

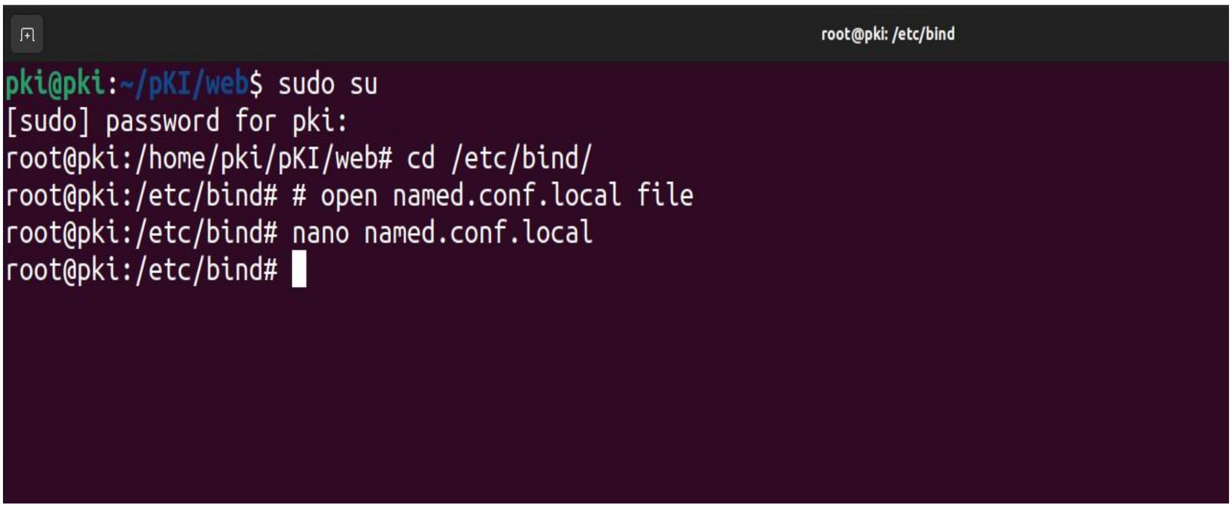
3) sudo apt install dnsutils

copy db.local to your db.shrutibhakta.local

4) sudo su

5) cd /etc/bind/

6) cp db.local db.shrutibhakta.local



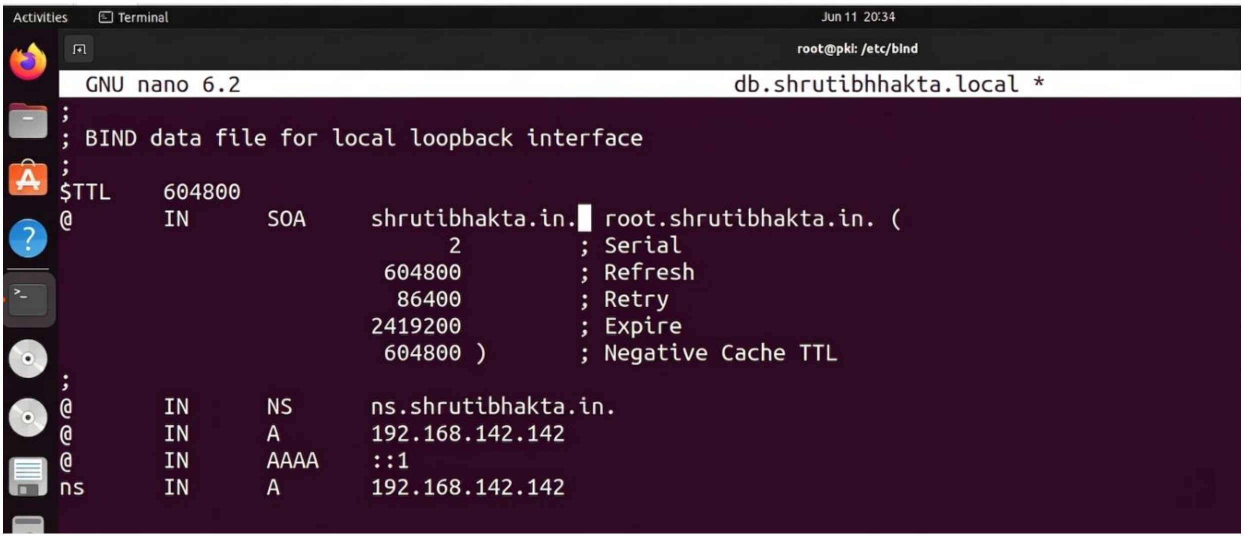
```
root@pki: /etc/bind
 GNU nano 6.2                                   named.conf.local *
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918"


zone "shrutibhakta.in" {
    type master;
    file "/etc/bind/db.shrutibhakta.local";
};
```

I have already bind9 installed so I will change the name of db.local file and edit

```
root@pki: /etc/bind
pki@pki:~/pki/web$ sudo su
[sudo] password for pki:
root@pki:/home/pki/pki/web# cd /etc/bind/
root@pki:/etc/bind# # open named.conf.local file
root@pki:/etc/bind# nano named.conf.local
root@pki:/etc/bind# nano named.conf.local
root@pki:/etc/bind# ls
bind.keys  db.127  db.ccee.com.local  db.local  named.conf.default-zones  named.conf.options  zones.rfc1918
db.0       db.255  db.empty           named.conf  named.conf.local        rndc.key
root@pki:/etc/bind# mv db.ccee.com.local db.shrutibhakta.local
root@pki:/etc/bind# ls
bind.keys  db.127  db.shrutibhakta.local db.local  named.conf.default-zones  named.conf.options  zones.rfc1918
db.0       db.255  db.empty              named.conf  named.conf.local       rndc.key
root@pki:/etc/bind#
```

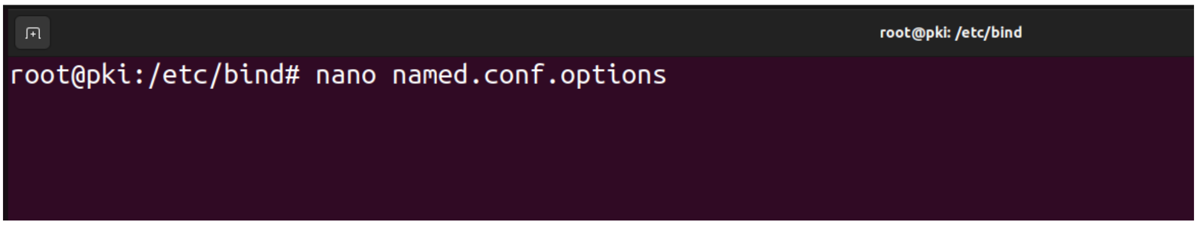
Now edit db.shrutibhakta.local file to create forward zone

```
root@pki: /etc/bind
root@pki:/e/tiod# nano db ob.sitrttibhakta.local
```



Save and exit

Now edit file named.conf.options



```
root@pki: /etc/bind
  GNU nano 6.2                                         named.conf.options
options {
        directory "/var/cache/bind";

        // If there is a firewall between you and nameservers you want
        // to talk to, you may need to fix the firewall to allow multiple
        // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

        // If your ISP provided one or more IP addresses for stable
        // nameservers, you probably want to use them as forwarders.
        // Uncomment the following block, and insert the addresses replacing
        // the all-0's placeholder.

         forwarders {
                192.168.142.142;
         };

        //========================================================================
        // If BIND logs error messages about the root key being expired,
        // you will need to update your keys.  See https://www.isc.org/bind-keys
        //========================================================================
        dnssec-validation auto;

        listen-on-v6 { any; };
```

Save and exit

Now restart bind 9 service

systemctl restart bind9.service

```
root@pki: /etc/bind
root@pki:/etc/bind# nano named.conf.options
root@pki:/etc/bind# systemctl restart bind9.service
root@pki:/etc/bind#
```

Now create a virtial host file but first copy wesite key, chain certificate and rootCA certificate in ~/certs folder

```
pki@pki: ~
pki@pki:~$ ls certs/
ccee.com.pem  ccee_key.pem  root_ca.crt
pki@pki:~$ #first remove all these data
pki@pki:~$ cd certs/
pki@pki:~/certs$ rm -rf *
pki@pki:~/certs$ ls
pki@pki:~/certs$ cd -
/home/pki
pki@pki:~$
```

```
pki@pki: ~
pki@pki:~$ ls certs/
shrutibhakta.key  fullchain.crt  rootCA11.crt
pki@pki:~$
```

Virtual host file

Go to sites-available folder

```
root@pki: /etc/apache2/sites-available
root@pki:/etc/bind# cd /etc/apache2/sites-available/
root@pki:/etc/apache2/sites-available#
```

```
root@pki: / etc/apache2/sites-available
root@pki:/etc/apache2/sites-available# ls
000-default.conf  ccee.com.conf  default-ssl.conf
root@pki:/etc/apache2/sites-available# # Now I need to rename file ccee.com.conf to
` shrutibhakta-ssl.conf
root@pki:/etc/apache2/sites-available# mv ccee.com.conf shrutibhakta-ssl.conf
```

Now open shrutibhakta-ssl.conf file

```
root@pki: /etc/apache2/sites-available
  GNU nano 6.2                shrutibhakta-ssl.conf *
<VirtualHost *:443>
    ServerName shrutibhakta.in
    DocumentRoot /var/www/html

    SSLEngine on
    SSLCertificateFile /home/pki/certs/fullchain.crt
    SSLCertificateKeyFile /home/pki/certs/shrutibhakta.key
    SSLCACertificateFile /home/pki/certs/rootCA11.crt

</VirtualHost>
```

Save and exit

Now enable the ssl module and enable the site

a2enmod ssl

a2ensite shrutibhakta-ssl

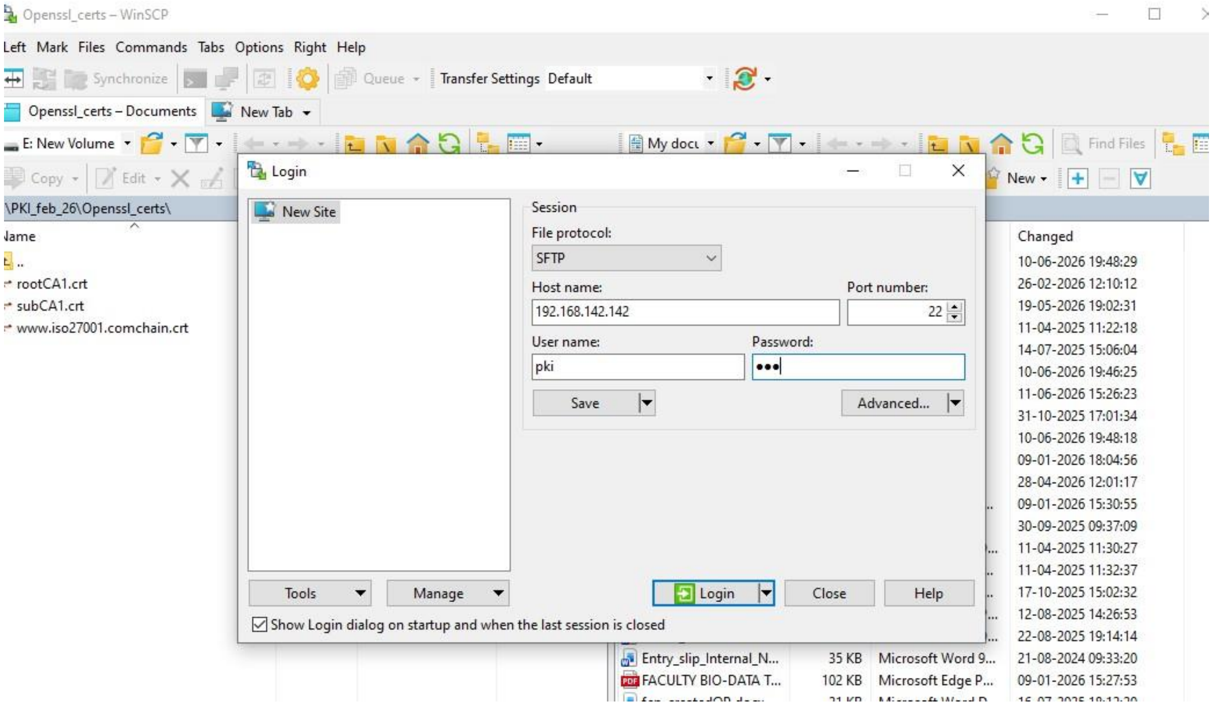
```
root@pki:/etc/apache2/sites-available
root@pki:/etc/apache2/sites-available# a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
root@pki:/etc/apache2/sites-available#
```

```
root@pki: /etc/apache2/sites-available
root@pki:/etc/apache2/sites-available# a2ensite shrutibhakta-ssl
Enabling site shrutibhakta-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@pki:/etc/apache2/sites-available#
```

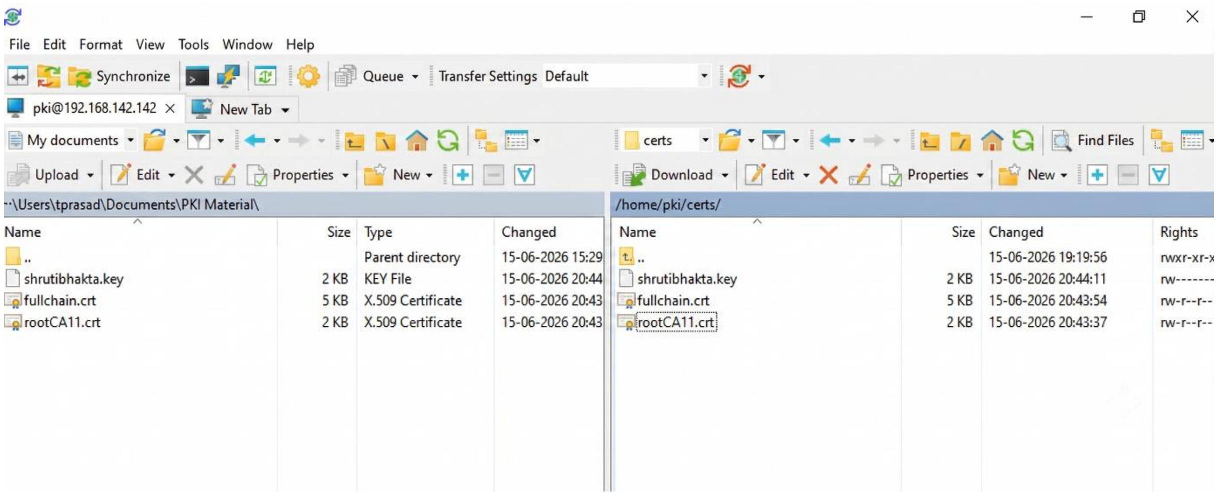
Now reload apache2 and restart

```
root@pki: /etc/apache2/sites-available
root@pki:/etc/apache2/sites-available# a2ensite shrutibhakta-ssl
Enabling site shrutibhakta-ssl.
Enabling site shrutibhakta-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@pki:/etc/apache2/sites-available# systemctl reload apache2
root@pki:/etc/apache2/sites-available# systemctl restart apache2
root@pki:/etc/apache2/sites-available#
```

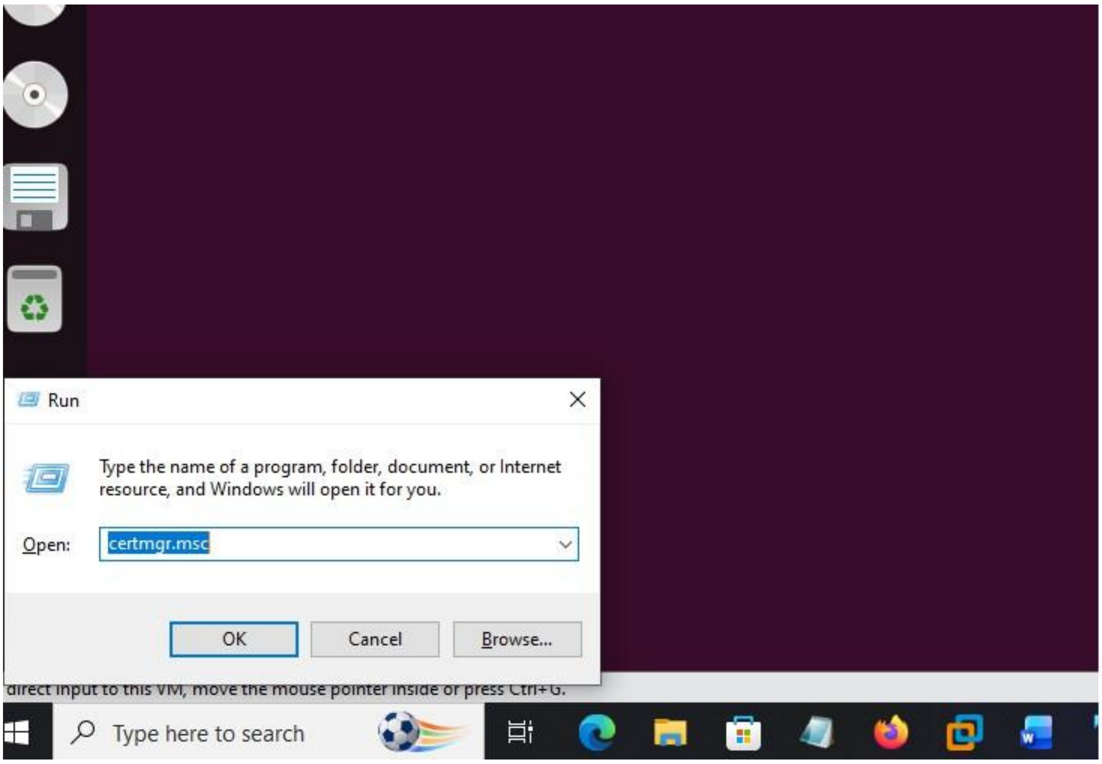
Now use winscp:

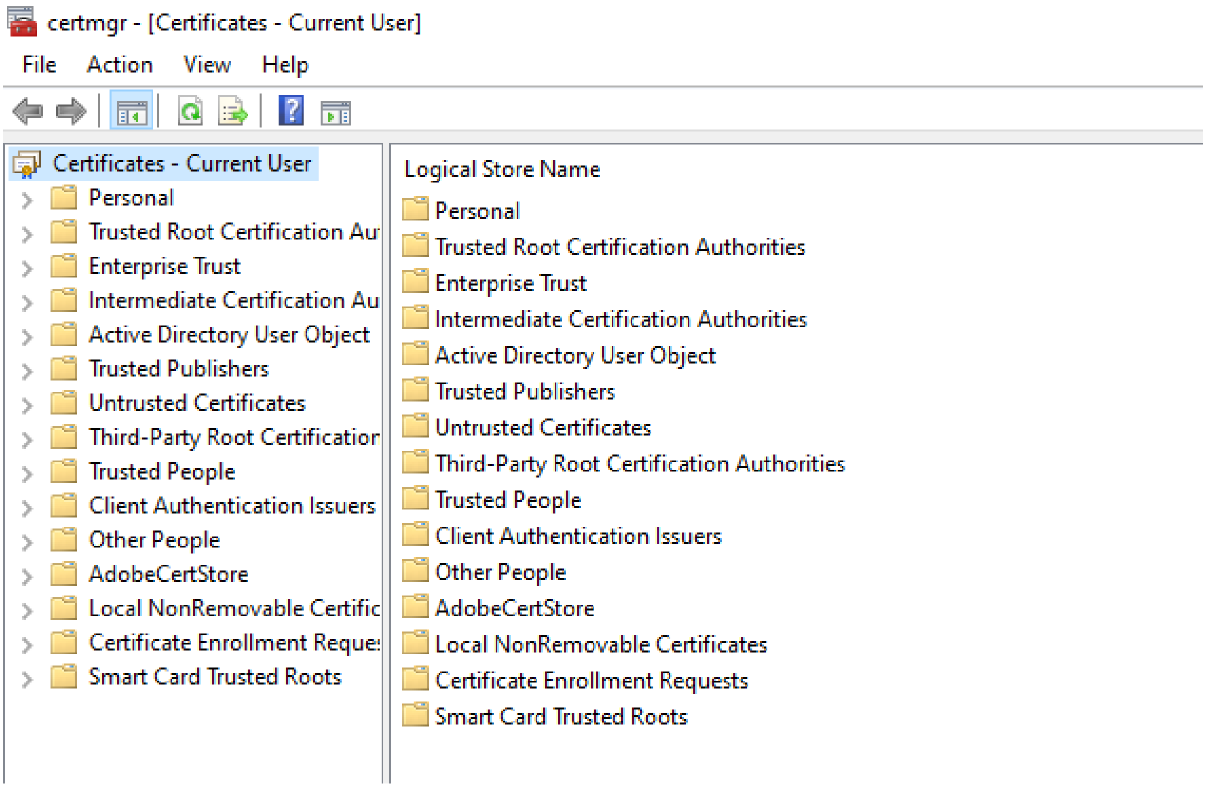


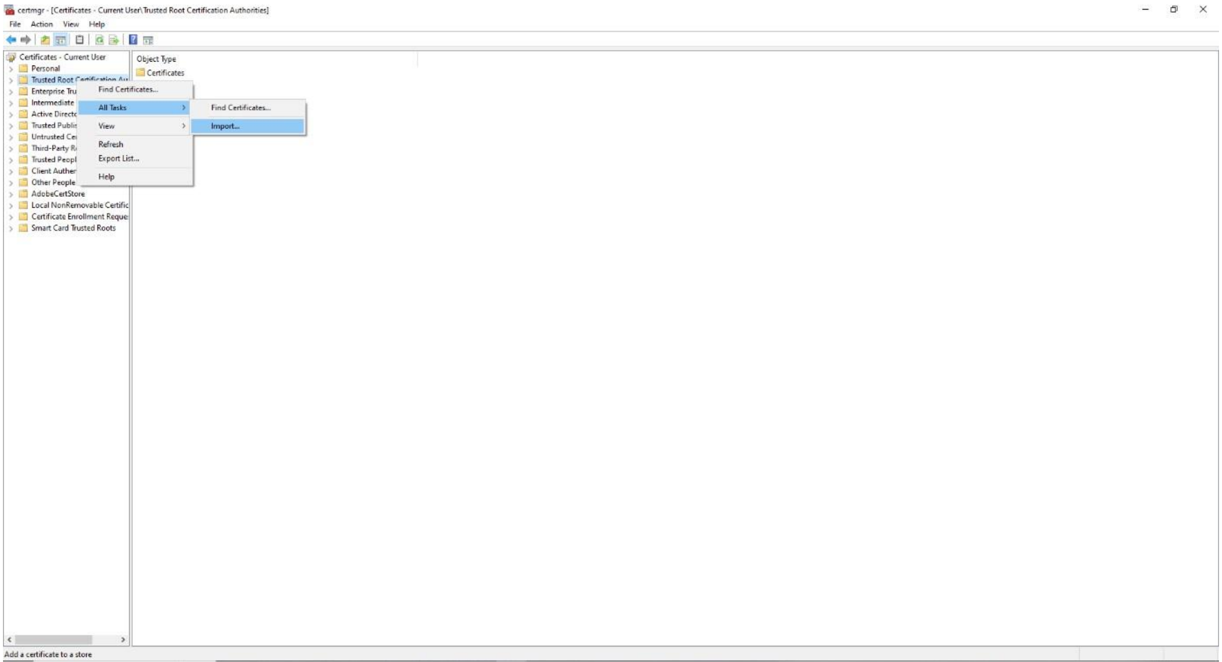
Copy all data of certs folder from ubuntu to base machine (windows 11)



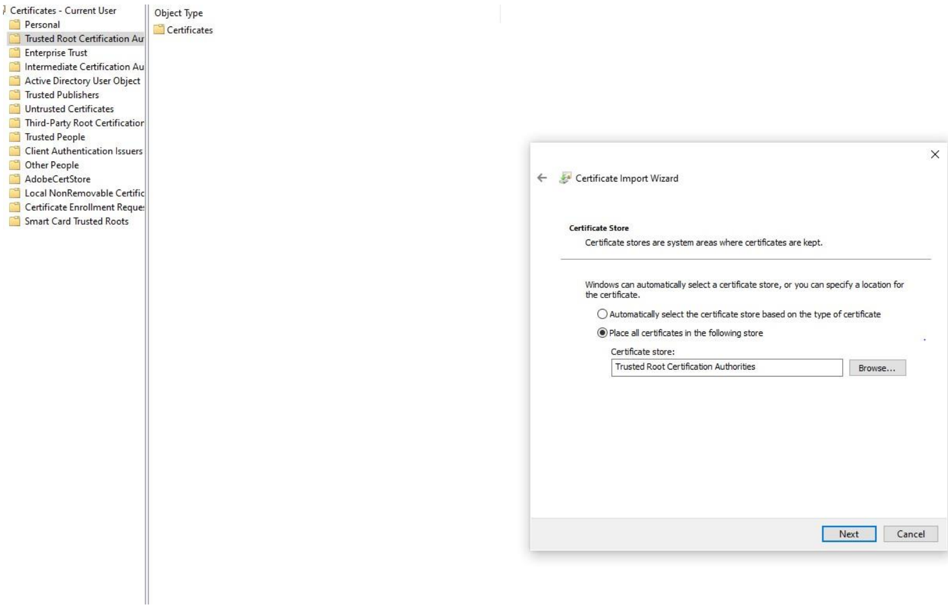
Import all certificates in cert manager

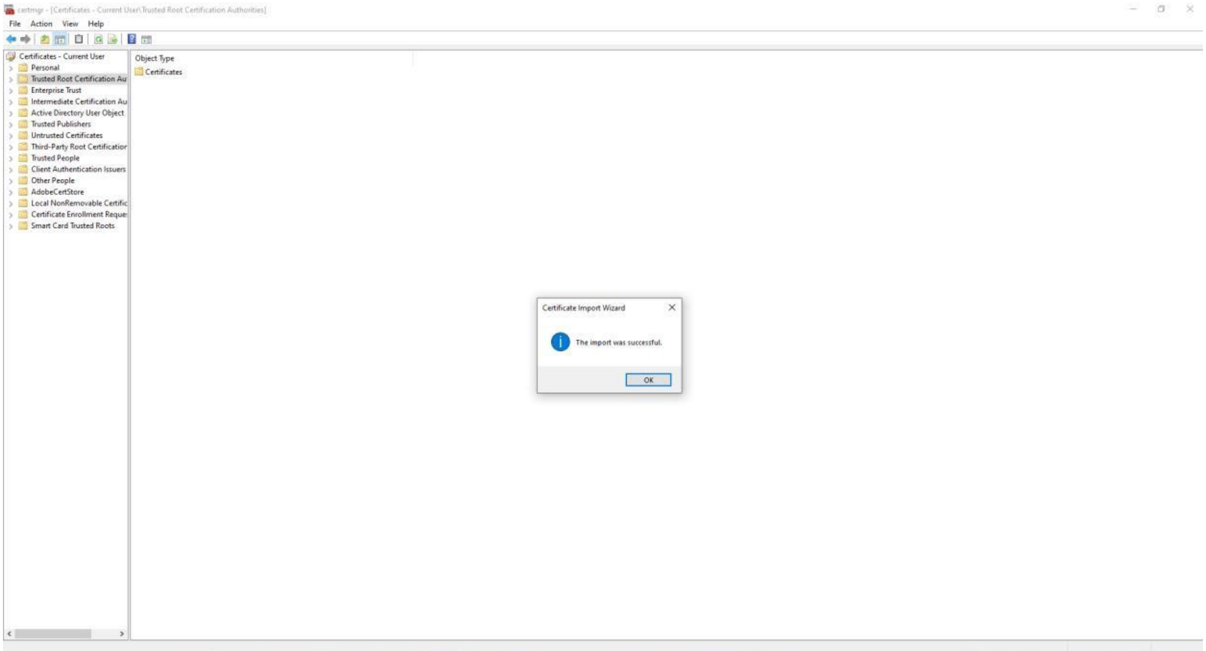

Run
Type the name of a program, folder, document, or Internet resource, and Windows will open it for you.
Open:
certmgr.msc
OK
Cancel
Browse...
direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search


certmgr - [Certificates - Current User]
File
Action
View
Help
Certificates - Current User
Personal
Trusted Root Certification Au
Enterprise Trust
Intermediate Certification Au
Active Directory User Object
Trusted Publishers
Untrusted Certificates
Third-Party Root Certificatior
Trusted People
Client Authentication Issuers
Other People
AdobeCertStore
Local NonRemovable Certific
Certificate Enrollment Reque:
Smart Card Trusted Roots
Logical Store Name
Personal
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Active Directory User Object
Trusted Publishers
Untrusted Certificates
Third-Party Root Certification Authorities
Trusted People
Client Authentication Issuers
Other People
AdobeCertStore
Local NonRemovable Certificates
Certificate Enrollment Requests
Smart Card Trusted Roots

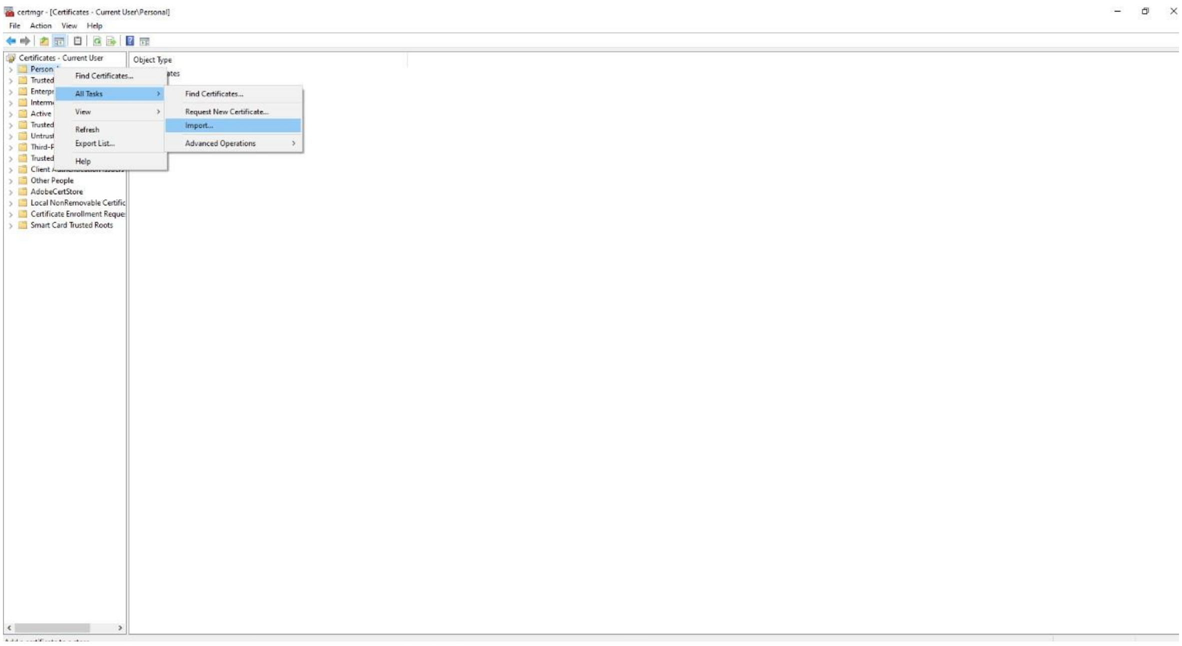


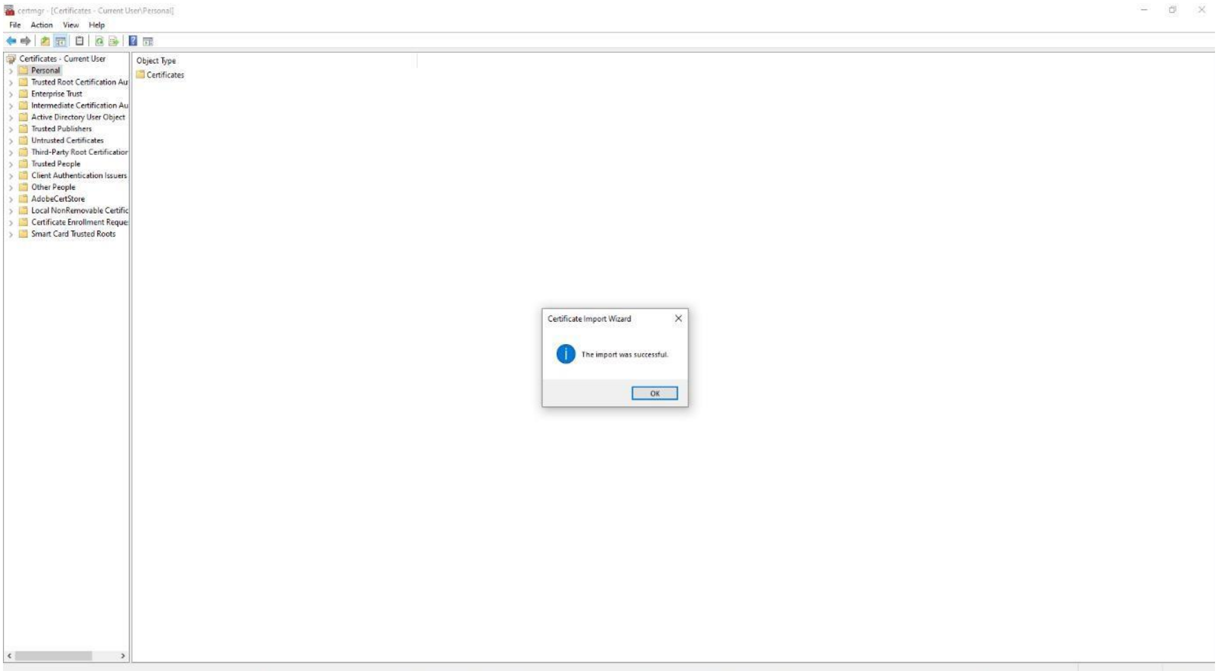
Import CA certificate





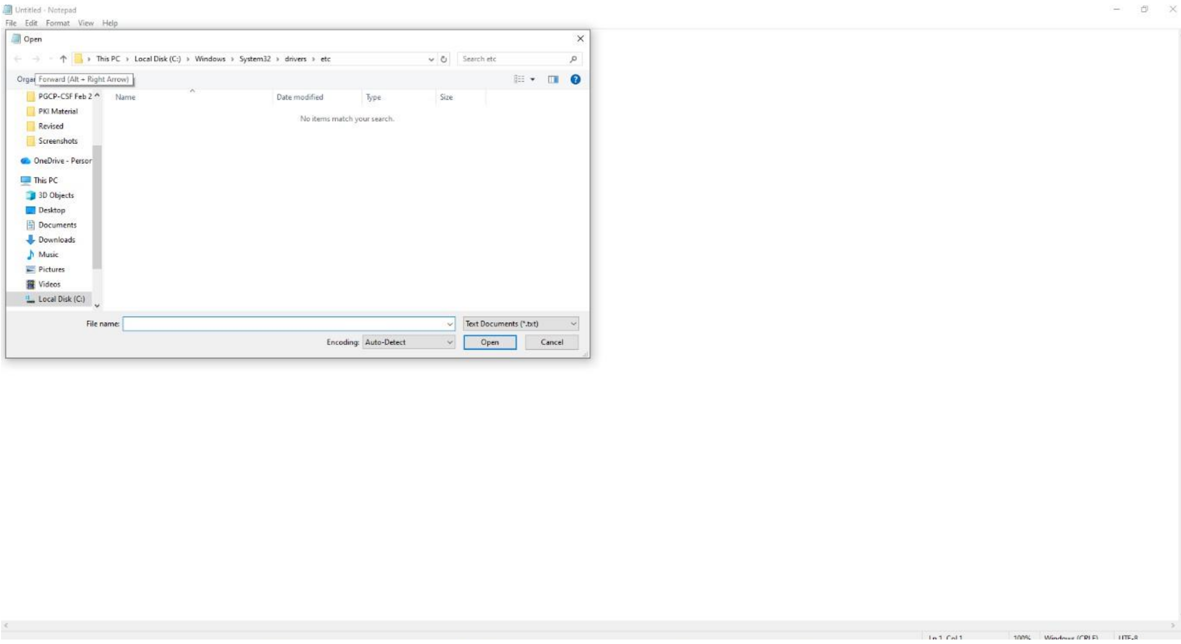
Import personal certificate



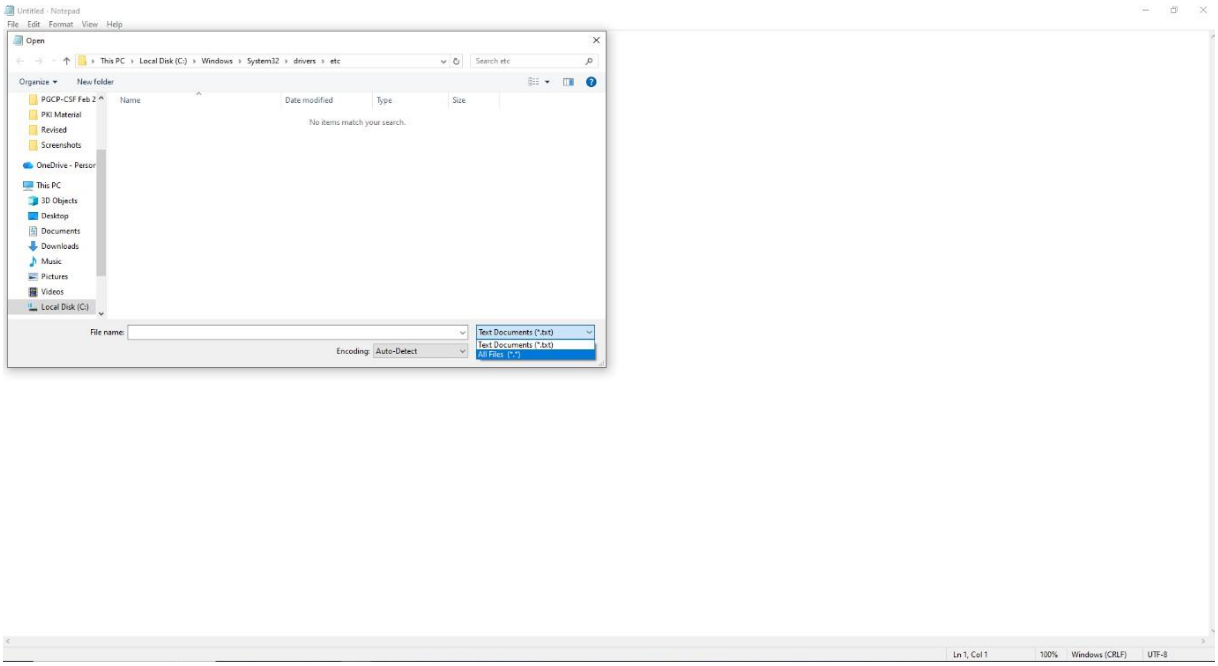


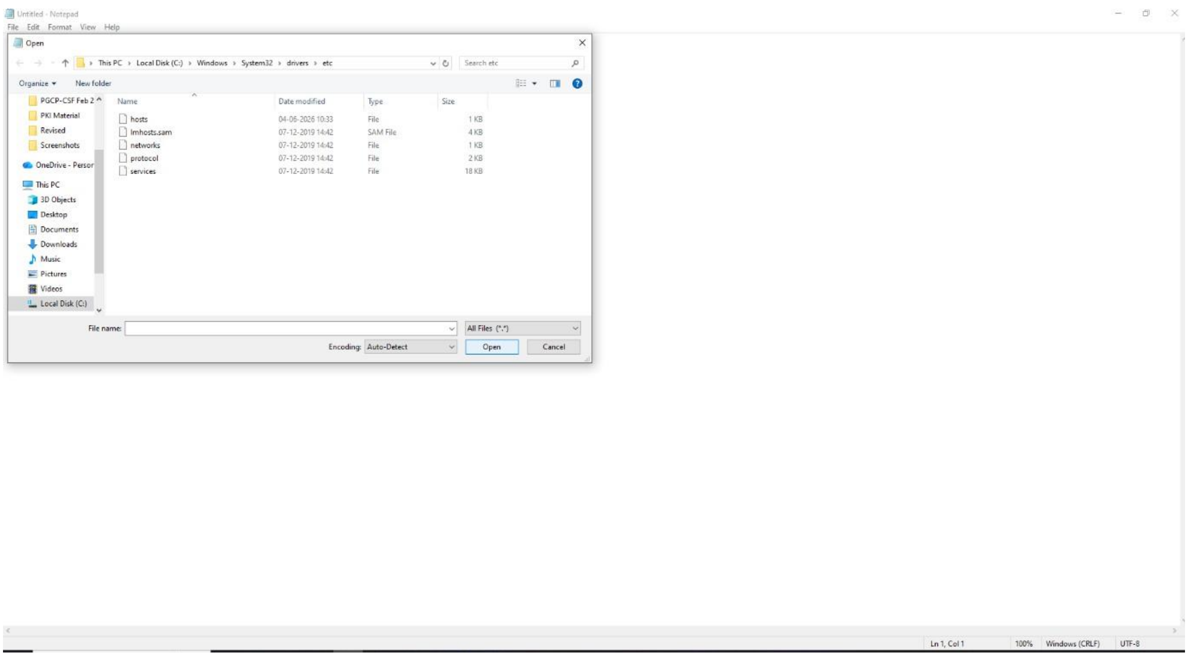
Edit your base machine host file

Right click on notepad and open as administrator



Internal


Untitled - Notepad
File Edit Format View Help
Open
This PC › Local Disk (C:) › Windows › System32 › drivers › etc
Search etc
Organize ▾ New folder
PGCP-CSF Feb 2
PKI Material
Revised
Screenshots
OneDrive - Persor
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Name
Date modified
Type
Size
No items match your search.
File name:
Encoding: Auto-Detect
Text Documents (*.txt)
Text Documents (*.txt)
All Files (*.*)
Ln 1, Col 1
100%
Windows (CRLF)
UTF-8


Untitled - Notepad
File Edit Format View Help
Open
This PC › Local Disk (C:) › Windows › System32 › drivers › etc
Search etc
Organize ▾ New folder
Name
Date modified
Type
Size
hosts
04-06-2026 10:33
File
1 KB
lmhosts.sam
07-12-2019 14:42
SAM File
4 KB
networks
07-12-2019 14:42
File
1 KB
protocol
07-12-2019 14:42
File
2 KB
services
07-12-2019 14:42
File
18 KB
File name:
All Files (*.*)
Encoding: Auto-Detect
Open
Cancel
Ln 1, Col 1
100%
Windows (CRLF)
UTF-8

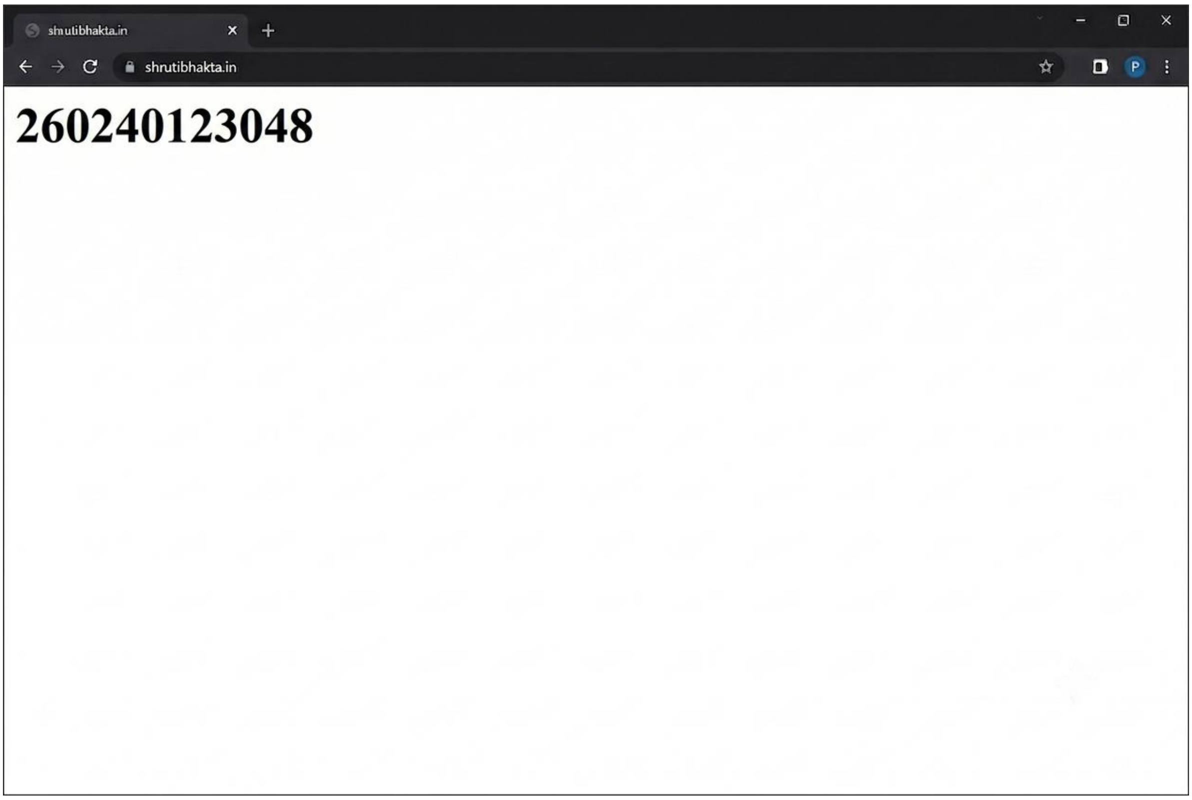
Now save the below file

*hosts - Notepad

File Edit Format View Help

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#      102.54.94.97     rhino.acme.com          # source server
#       38.25.63.10     x.acme.com              # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1       localhost
#       ::1             localhost

192.168.142.142         www.shrutibhakta.in
```

Save and exit then check into browser in private mode or normal mode


shrutibhakta.in
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